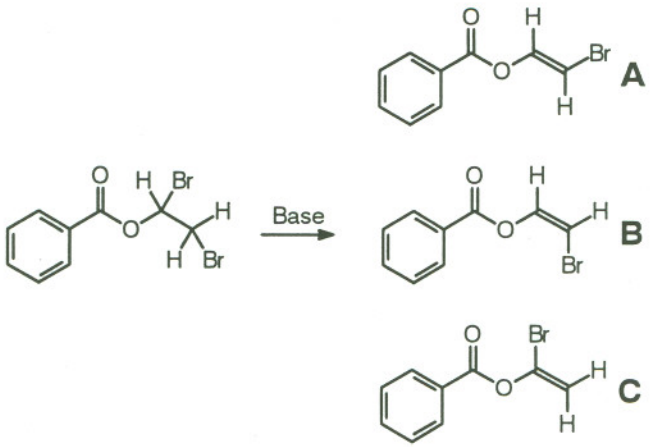
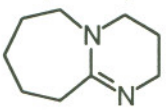
1. *JACS* **1994**, *116*, 789

Treatment of the dibromobenzoate ester below with base in nonaqueous solvents (THF or toluene) induces elimination of HBr, presumably by an E2 mechanism.

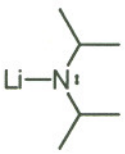


As you can see, there are three possible isomeric products. When an "ordinary" base, such as DBU, is used for the reaction, all three are produced in large yields. But when a metal amide, such as LDA, is the base, **C** is produced to a much greater extent than **A** or **B**.
Provide an explanation for this result.

DBU = 1,8-diazabicyclo[5.4.0]undec-7-ene. This is probably not important.

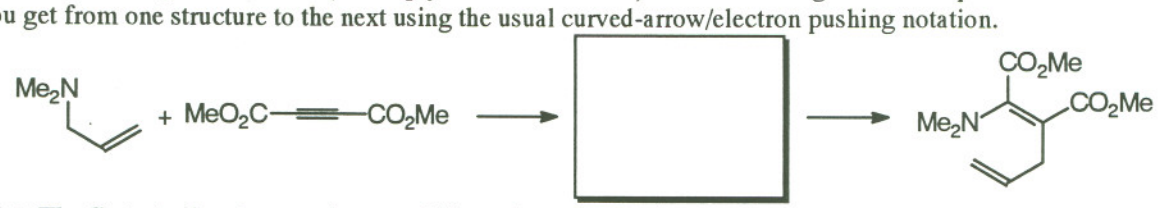


LDA = lithium diisopropyl amide. This is important.



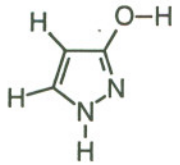
2. *JACS* **1994**, *116*, 579

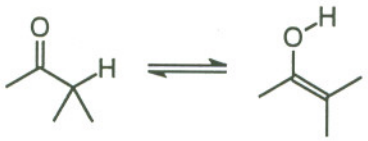
Fill in the missing intermediate (the empty box in the middle) in the following reaction sequence. Show how you get from one structure to the next using the usual curved-arrow/electron pushing notation.



Hint: The first step involves conjugate addition; the second a sigmatropic rearrangement.

3. Shown below is one of the tautomers of 3-hydroxypyrazole.



Draw <u>all</u> possible tautomers of this molecule. You may assume the five-membered ring is planar in all the tautomeric forms.

Reminder: tautomerism is a special case of isomerism which involve <u>distinct</u> chemical species that can rapidly interconvert from one to another. The most familiar kind is the keto-enol tautomerism in carbonyl compounds: